UNIVERSITY OF GHANA

(All rights reserved)

DEPARTMENT OF TEACHER EDUCATION
SCHOOL OF EDUCATION AND LEADERSHIP
COLLEGE OF EDUCATION
END OF YEAR TWO SEMESTER TWO EXAMINATIONS, 2024/2025
B.ED. PROGRAMME

COURSE CODE: TEJS 254
COURSE TITLE: MAP USE AND INTERPRETATION

Instruction: Answer **all** questions in Section A and any **three** in Section B. **All questions must be answered in the answer booklet.**

Time: 2 hours

SECTION A

[25 Marks]

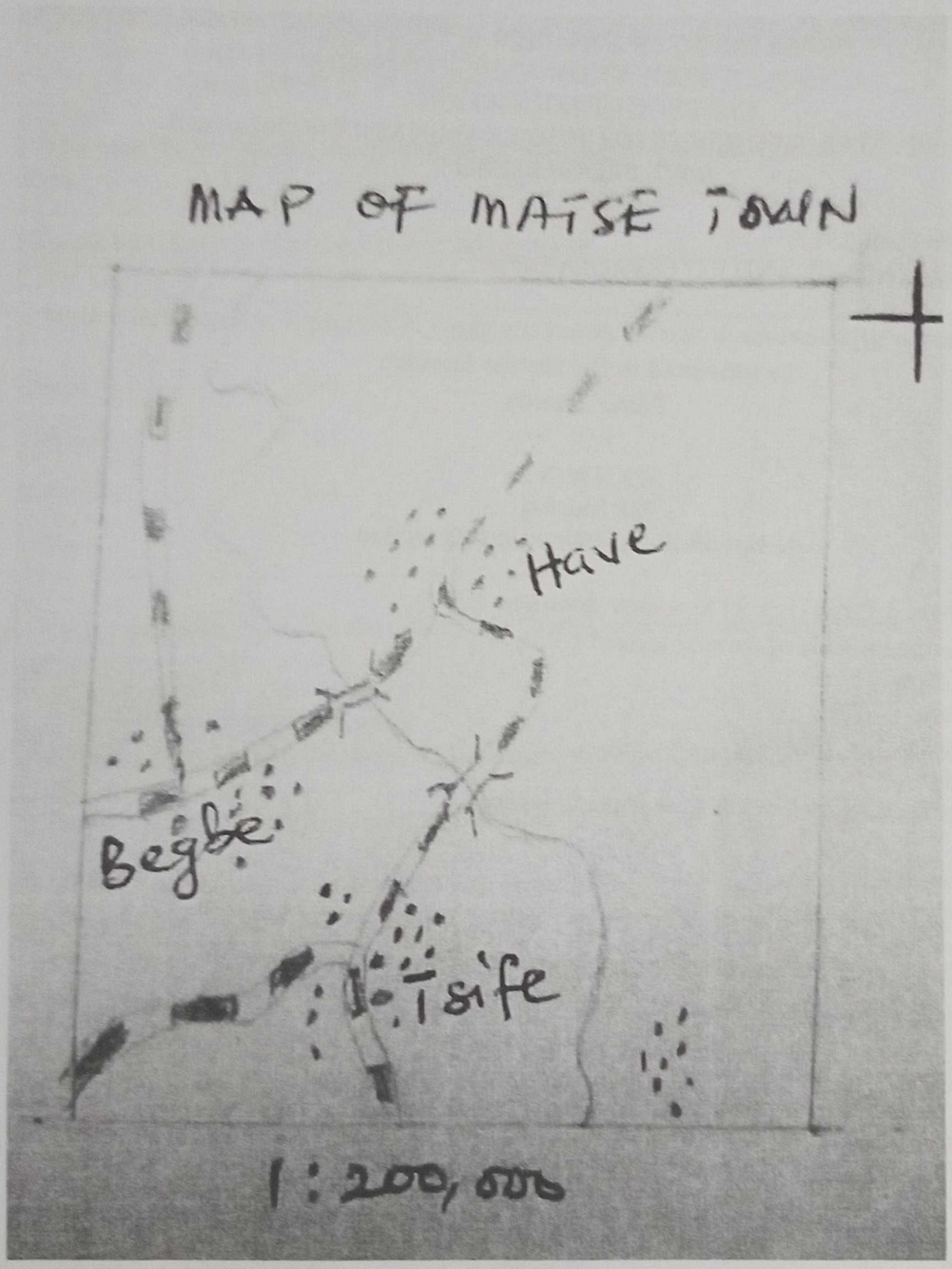
Attempt all the questions in this section.

Use the map of Matse Town (Fig. 1) to answer questions 1-5

1. What is the statement scale of the map in km?
   A. 1cm to 0.2km
   B. 1cm to 1km
   C. 1cm to 20km
   D. 1cm to 2km
2. What is the distance by road in km from Have to Begbe?
   A. 4.5km
   B. 3.2km
   C. 6.6km
   D. 5.2km
3. What is the bearing of Tsife from Begbe using the whole circle bearing?
   A. 310°
   B. 130°
   C. 270°
   D. 180°
4. What would the new scale of Matse Town map if it is drawn to a third of its size?
   A. 1:600,000
   B. 1:400,000
   C. 1:300,000
   D. 1:500,000

5. What is the direction of Begbe from Have?
   A. Northeast
   B. Southeast
   C. Northwest
   D. Southwest

Fig. 1: Map of Matse Town



6. "The bearing of a mountain from here is 270°". What is the location of the mountain from the observer?
   A. North
   B. South
   C. East
   D. West
7. Which type of survey station is primarily used in triangulation to determine the position of points through angles?
   A. Trigonometrical station
   B. Spot height station
   C. Contour station
   D. Form line station
8. Which statistical map best shows population size of cities?
   A. Choropleth
   B. Dot map
   C. Proportional symbol map
   D. Flow map
9. The distance *"as the crow flies"* from A to B is 5.5cm on a map. Calculate the ground distance in kilometres using the scale 1:50,000.
   A. 2.8 km
   B. 2.6 km
   C. 2.9 km
   D. 2.5 km
10. If a bearing is given as 045°, what is the direction relative to the north?
   A. North-East
   B. South-East
   C. South-West
   D. North-West
11. What is primarily represented in cadastral surveying?
   A. Elevation of lands
   B. Land boundary details
   C. Road transportation routes
   D Natural features
12. Which source of data is most commonly used in creating modern maps?
   A. Legends
   B. Satellite imagery
   C. Ancient scrolls
   D. Hand-drawn sketches
13. Which of these scales is considered to be for a small-scale map?
   A. 1:10,000
   B. 1:25,000
   C. 1:250,000
   D. 1:50,000

14. How would you express the statement scale: *one centimeter to five kilometres* in representative fraction?
   A. 1: 500
   B. 1: 5000
   C. 1: 50000
   D. 1: 500000
15. Contour lines that are close together indicate:
   A. Gentle slope
   B. Spur
   C. Steep valley
   D. Steep slope
16. The grid reference system is used to:
   A. Locate places on a map
   B. Indicate time zones
   C. Show vegetation
   D. Show altitude
17. Which type of map shows land ownership and boundaries?
   A. Topographic map
   B. Political map
   C. Thematic map
   D. Cadastral map
18. A thematic map focuses on:
   A. Elevation
   B. Roads and railways
   C. Specific themes like climate or population
   D. National boundaries
19. What is the primary reason cartographers rely on multiple sources, such as aerial photos, remote sensing, and field notes when compiling maps?
   A. To reduce the cost and duration of the printing process
   B. To follow traditional and religious mapping conventions
   C. To improve the precision and reliability of geographic data
   D. To comply with the international grid system requirements
20. If a civil engineer wants to study landforms and human-made features like bridges, which type of map would be most suitable?
   A. Political map of towns and districts
   B. Population map showing regional densities
   C. Soil map showing composition and fertility
   D. Topographic map showing relief and infrastructure
21. Convert the statement scale of 4cm to 2km into representative fraction (RF) scale.
   A. 1:80,000
   B. 1:20,000
   C. 1:50,000
   D. 1:100,000

22. Which direction is based on the geographic axis around which the Earth rotates and remains fixed regardless of map type or location?
   A. Grid North
   B. Magnetic North
   C. North-East
   D. True North
23. If the bearing from Location X to Location Y is 070°, what is the back bearing from Y to X?
   A. 070°
   B. 180°
   C. 250°
   D. 290°
24.. A place is located at 30°E and another at 60°E. If it's 12:00pm at the second place, what time is it at the first?
   A. 10:00am
   B. 12:00pm
   C. 1:00pm
   D. 2:00pm
25. What is the local time difference between two towns 60° apart in longitude?
   A. 1 hour
   B. 2 hours
   C. 4 hours
   D. 6 hours

SECTION B

[75 Marks]

Answer Question One (1) and any other two (2) questions from this section. Each question carries 25 marks.

1. Use the description below to answer all the questions numbered (a) to (e) in question one. Show all workings where applicable.

A district is represented on a topographical map with the following specifications:

i. Map Scale: 1:100,000,
ii. Original Map Dimensions: 20 cm (length) × 16 cm (width),
iii. A River named Kokum meanders from the northeast to the southwest, passing near the southern boundary of the map. The measured length of the river on the original map is 12 cm,
iv. The settlement of Adom and a mining site are located adjacent to the River Kokum, in the southern part of the district,
v. You are required to reduce the map to half (½) of its original size and interpret accordingly.

a) Calculate the actual ground area (in square kilometers) covered by the original map. [4 marks]
b) Draw a rectangular outline of the reduced map using accurate dimensions. [6 marks]
c) Calculate the new scale of the reduced map. [4 marks]

d) Using the original map scale, calculate the actual ground length of River Kokum. [2 marks]
e) Identify and explain four possible land-use functions of the mapped area. [4 marks]

2. a) Explain three major types of bearing. (9marks)
b) With a suitable scale of Northings and Eastings, draw a grid system and locate the following towns using the attached figure references;
i. Matse (2345) (4 marks)
ii. Kpedze (2140) (4 marks)
iii. Klefe (223427) (4 marks)
iv. Ziavi (245412) (4 marks)

3. The table below shows the quantities of various commodities exported by Ghana in 2022 and 2024.

Exports by Ghana in 1980 and 1990 ('000 tonnes)

Commodities	1980	1990
Coffee	360	640
Cocoa	560	920
Timber	206	480
Kolanuts	188	420
Others	102	300

a. Using the table, calculate the percentage change in commodity export between 1980 and 1990 for:
i. Cocoa [3 marks]
ii. Total commodity export [4 marks]
b. Represent the data by means of two group bars graphs using the graph paper provided. [10 marks]
c. State two advantages and two disadvantages of group bar graph. [2 marks × 4 = 8 marks]

4. With the aid of well-labelled diagrams, describe the following relief features and drainage patterns as they appear on topographic maps:
(i) Conical hill
(ii) Valley
(iii) Escarpment
(iv) Dendritic drainage pattern
(v) Trellis drainage pattern. [Each feature carries 4 marks]

5. Discuss how symbols and keys are used on maps to represent physical and cultural features, and assess their importance in map interpretation.